

NIFTY100 Financial Intelligence Platform

Analyst Guide

Financial Analytics Platform

Technologies

- Python
- Streamlit
- FastAPI
- SQLite
- Plotly
- Pandas

Document Purpose

This guide provides complete instructions for analysts, investors, students, and developers to effectively use the NIFTY100 Financial Intelligence Platform. It covers installation, dashboard navigation, financial screening, PDF report generation, REST API usage, troubleshooting, and best practices.

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Introduction

About the Platform

The NIFTY100 Financial Intelligence Platform is an end-to-end financial analysis application designed to simplify the evaluation of publicly listed Indian companies. It integrates financial statement analysis, stock screening, portfolio analytics, valuation models, and interactive dashboards into a single platform.

The application automates the process of importing financial data, validating datasets, calculating financial ratios, and generating professional investment reports. It enables analysts to make informed investment decisions through data-driven insights and interactive visualizations.

Major Capabilities

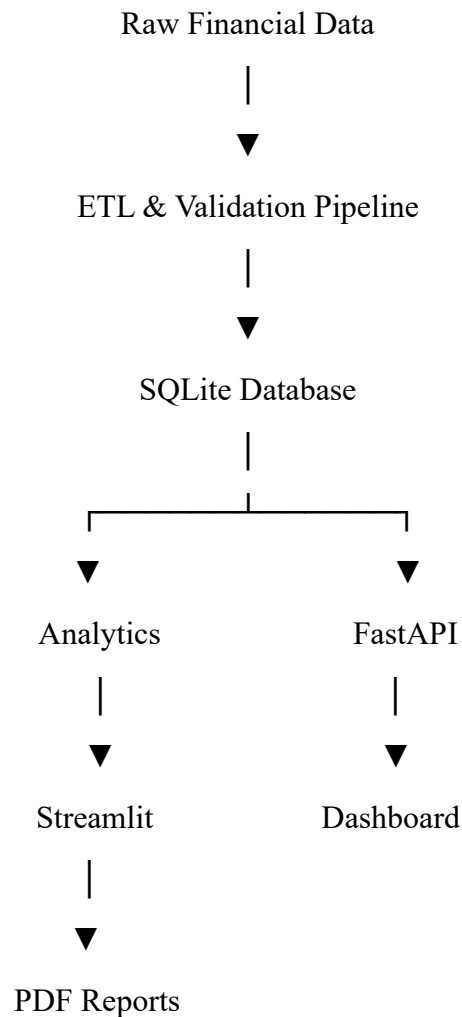
- Automated ETL pipeline
- Financial statement analysis
- Interactive stock screener
- Company comparison
- Portfolio analytics
- Sector-wise analysis
- PDF tearsheet generation
- REST API
- SQLite database backend

Intended Users

- Financial Analysts
- Investors
- Students
- Researchers
- Portfolio Managers
- Developers

System Architecture

The platform follows a modular architecture consisting of data ingestion, database management, analytics, reporting, and visualization components.



Components

ETL Layer

Responsible for:

- Reading Excel files
- Cleaning data
- Validation
- Normalization
- Database loading

Database Layer

Stores

- Companies
- Financial statements
- Ratios
- Stock prices
- Peer groups
- Sectors
- NLP insights

Analytics Layer

Calculates

- CAGR
- Growth rates
- Financial ratios
- Valuation metrics
- Portfolio statistics

Presentation Layer

Provides

- Streamlit Dashboard
- REST API
- PDF Reports

Installation & Setup

This section should explain:

Prerequisites

- Python 3.12+
- Git
- SQLite

Installation

```
git clone <repository>
```

```
cd nifty100-project
```

```
python -m venv .venv
```

```
pip install -r requirements.txt
```

Load Database

```
python src/etl/load_data.py
```

Launch Dashboard

```
streamlit run src/dashboard/app.py
```

Launch API

```
uvicorn src.api.main:app --reload
```

Include a screenshot of:

- Terminal after ETL
- Streamlit home page

Streamlit Dashboard

The Streamlit Dashboard serves as the primary user interface of the NIFTY100 Financial Intelligence Platform. It enables analysts to interact with financial data through an intuitive, web-based environment without requiring SQL queries or programming knowledge.

After launching the application using:

```
streamlit run src/dashboard/app.py
```

the dashboard becomes available at:

<http://localhost:8501>

Dashboard Layout

The interface consists of three primary sections:

1. Sidebar Navigation

The left sidebar provides access to all available modules.

Typical navigation items include:

- Home
- Company Profile
- Cluster Analytics
- Trend Analysis
- Stock Screener
- Sector Analysis
- Peer Comparison
- Capital Allocation
- Rankings
- Portfolio
- Reports

The sidebar remains visible throughout the session, allowing users to switch between modules without restarting the application.

2. Main Workspace

The center panel displays interactive content based on the selected module.

Depending on the current page, it may contain:

- KPI cards

- Interactive charts
- Financial tables
- Company profiles
- Screening results
- Investment insights

3. Filters Panel

Most pages include filters that allow analysts to customize the displayed data.

Common filters include:

- Company Name
- Sector
- Market Capitalization
- Financial Year
- Revenue Growth
- ROE
- Debt-to-Equity Ratio
- P/E Ratio

Changing any filter automatically refreshes the visualizations and tables.

Dashboard Features

The dashboard provides several interactive capabilities:

- Dynamic filtering
- Interactive Plotly charts
- Downloadable tables
- Real-time KPI updates
- Company comparison
- PDF report generation
- Responsive layouts

Navigation

- Home
- Company Profile
- Cluster Analytics
- Trend Analysis
- Stock Screener
- Sector Analysis
- Peer Comparison
- Capital Allocation
- Rankings
- Portfolio
- Reports

 Database

Companies
92

Sectors

10

Platform

ML Model

K-Means

Analytics

Showing Dashboard for Sep 2024

Companies
1

Average ROE

17.70%

Median P/E

33.32

Median D/E

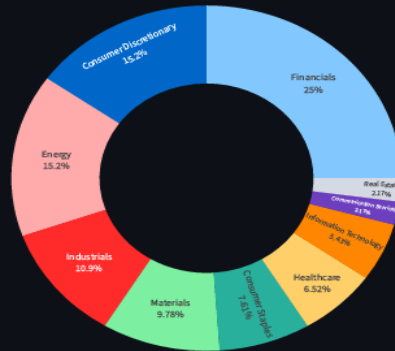
0.02

Revenue CAGR
11.19%

Debt Free
1

Sector Distribution

Companies by Broad Sector



🏆 Top 5 Companies by Composite Quality Score

Financial Screener

The Financial Screener allows users to identify companies that satisfy specific investment criteria. Instead of manually reviewing every company, analysts can quickly filter organizations based on predefined financial metrics.

The screener queries the SQLite database and returns matching companies instantly.

Accessing the Screener

From the navigation sidebar:

Home



Financial Screener

Available Filters

Typical screening parameters include:

Filter	Description
Sector	Industry category
Market Capitalization	Company size
Revenue Growth	Annual revenue growth
Profit Growth	Net profit growth
ROE	Return on Equity
ROCE	Return on Capital Employed
Debt-to-Equity	Financial leverage
Current Ratio	Liquidity
P/E Ratio	Valuation
P/B Ratio	Book valuation

Example Screening Process

Suppose an analyst wants companies with:

- ROE greater than 20%

- Debt-to-Equity less than 1
- P/E Ratio below 30

Configure the filters as follows:

$ROE > 20$

$Debt/Equity < 1$

$P/E < 30$

Click **Apply Filters**.

The screener displays only companies matching these conditions.

Results Table

The results typically include:

- Company Name
- Sector
- Market Capitalization
- Revenue
- Net Profit
- ROE
- ROCE
- P/E Ratio
- Recommendation Score

Results can be exported for further analysis.

Best Practices

- Avoid applying too many filters simultaneously.
- Review companies individually after screening.
- Compare screening results across sectors.
- Combine financial metrics with qualitative analysis before making investment decisions.

15.00

10.00

10.00

15.00

50.00

5.00

1.00

5.00

0.00

Reset Filters

 [Download Filtered Companies \(CSV\)](#)

 Best Match

Company: Adani Ports & Special Economic Zone Ltd

Company Analysis

The Company Analysis page provides detailed financial information for an individual organization.

Users select a company from the dropdown list, after which the dashboard automatically retrieves all available financial information from the database.

Company Selection

Use the dropdown menu to select a company.

Example:

Select Company



TCS

Information Displayed

The dashboard presents several categories of information.

Company Overview

Displays:

- Company Name
- Industry
- Market Capitalization
- Business Description

Financial Statements

Interactive tables include:

- Profit & Loss Statement
- Balance Sheet
- Cash Flow Statement

Financial Ratios

Key ratios displayed include:

- ROE
- ROCE
- Debt-to-Equity
- Current Ratio
- Operating Margin
- Net Profit Margin
- EPS
- Dividend Yield

Interactive Charts

Typical visualizations include:

- Revenue Trend
- Profit Growth
- Cash Flow Trend
- Share Price History
- Margin Analysis

Charts support zooming, panning, and hover-based tooltips for detailed exploration.

Investment Summary

The platform automatically generates a concise summary containing:

Pros

- Strong revenue growth
- Healthy return ratios
- Stable cash flow

Cons

- High valuation
- Declining margins
- Elevated debt levels

These insights help analysts quickly identify the strengths and weaknesses of the selected company.

Peer Comparison

Users can compare the selected company with other organizations in the same sector.

Comparison metrics include:

- Revenue
- Profit
- ROE
- ROCE
- Market Capitalization
- P/E Ratio

This facilitates benchmarking against industry competitors.

Portfolio Analytics

The Portfolio Analytics module helps investors evaluate the performance and risk characteristics of a portfolio containing multiple NIFTY100 companies. Instead of analyzing stocks individually, this module provides an overall view of portfolio diversification, allocation, and expected returns.

The dashboard uses financial metrics, historical price data, and portfolio weights to generate meaningful investment insights.

Accessing Portfolio Analytics

From the sidebar navigation:

Home



Portfolio Analytics

Portfolio Configuration

Users can create a custom portfolio by selecting one or more companies and assigning investment weights.

Example:

Company	Allocation
TCS	30%
Infosys	25%
Reliance Industries	20%
HDFC Bank	15%
Larsen & Toubro	10%

The total allocation must equal **100%**.

Portfolio Metrics

The dashboard automatically calculates important portfolio statistics, including:

- Total Investment Value
- Expected Annual Return

- Portfolio Risk
- Sector Allocation
- Diversification Score
- Average P/E Ratio
- Average ROE
- Portfolio CAGR

These metrics help analysts evaluate whether the portfolio aligns with their investment objectives.

Portfolio Visualizations

Interactive charts available in this module include:

- Portfolio Allocation Pie Chart
- Sector Distribution
- Portfolio Growth Trend
- Risk vs. Return Scatter Plot
- Company Weight Distribution
- Historical Performance Comparison

Users can hover over chart elements to view detailed values and use built-in Plotly controls for zooming and exporting charts.

Portfolio Recommendations

Based on portfolio composition, the system provides automated suggestions such as:

- Reduce exposure to a single sector.
- Increase diversification across industries.
- Consider companies with stronger cash flow.
- Review holdings with consistently low ROE.
- Rebalance allocations periodically.

These recommendations are intended to support investment analysis and should be combined with independent financial research.



PDF Tearsheet Generation

The platform can generate professional PDF reports for individual companies. A tearsheet consolidates important financial information into a concise, printable report suitable for investment research and presentations.

Each tearsheet is generated automatically from the latest data stored in the SQLite database.

Purpose

The PDF tearsheet provides a comprehensive summary of a company's financial health, performance, and valuation in a single document.

Typical users include:

- Financial Analysts
- Investment Advisors
- Students
- Researchers
- Portfolio Managers

Generating a Tearsheet

Run the following command:

```
python src/reports/tearsheet_generator.py
```

If your project generates reports from the dashboard, users can also click the **Generate PDF** button after selecting a company.

Output Location

Generated reports are saved in:

output/

└─ tearsheets/

Example:

output/

└─ tearsheets/

|— TCS.pdf
|— INFY.pdf
|— RELIANCE.pdf
└— HDFCBANK.pdf

Contents of a Tearsheet

Each PDF report typically includes:

- Company Overview
- Business Summary
- Revenue Trend
- Profit Trend
- Cash Flow Summary
- Financial Ratios
- Historical Stock Performance
- Valuation Metrics
- Investment Pros
- Investment Cons
- Overall Recommendation

This standardized format enables quick comparison between companies.

Adani Energy Solutions Ltd

Investment Rating : Average

Company Overview

Website	https://www.adanienergysolutions.com/
Sector	nan
Market Cap	348,517.36 Cr
PE Ratio	58.5
PB Ratio	9.05
Dividend Yield	1.65 %

Financial Quality

ROE	8.59
ROCE	9.0
CFO Quality	High Quality
FCF CAGR	nan%
Capital Pattern	Excellent

Capital Allocation

Capital Score	63.0
Reinvestment	Growth Focused
Debt Trend	Increasing Debt
Leverage	Very High

Business Overview

AESL, part of the Adani portfolio, is a multidimensional organization with presence in various facets of the energy domain, namely power transmission, distribution, smart metering, and cooling solutions. AESL is India's largest private transmission company

Strengths

Strong free cash flow generation over 5 years signals healthy business fundamentals.
Revenue growing above 15% CAGR over 5 years reflects strong business momentum.
Operating profit margin above 25% indicates strong pricing power and cost discipline.

Risks

Debt-to-equity ratio of 2.93 is elevated for a non-financial company and warrants monitoring.
Rising debt over the past 3 years suggests increasing financial leverage risk.
Earnings per share declining for 3 consecutive years reflects deteriorating profitability.

Best Practices

- Refresh the database before generating reports.
- Verify that all ETL processes have completed successfully.

- Regenerate tearsheets after updating financial data.
- Store reports in the designated output directory for easy retrieval.